W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

PENETRATION TESTING REPORT

FOOTPRINTING, OSINT RECONNAISSANCE & NETWORK SCANNING PHASES

Engagement Field	Project Details
Pentester Name (Cybersecurity Professional)	Portia Dhladla (Mimie)
Program / Batch	BO84 NETWORKWALKS
Date	10 November 2026
Modules Completed	1. Open Source Intelligence (OSINT) & Domain Reconnaissance via theHarvester 2. Local Network Discovery & Scanning via Zenmap
Client / Target	1. Networkwalks / microsoft.com (Authorized Educational Scope) 2. My own local LAN Network (10.0.0.0/24)
Permission Secured from Client?	Yes
Phases Covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3–5: In Progress

1. Liability Disclaimer

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2. Introduction

This report covers the footprinting and reconnaissance phase targeting the microsoft.com domain using theHarvester in Kali Linux, alongside active network scanning on my own local LAN network using Zenmap. Together, these two modules demonstrate how an attacker or security professional moves from gathering publicly available external information to actively mapping live hosts on a local network.

This assessment is part of Week 2 of my ongoing Cybersecurity internship program at Networkwalks (Batch BO84). Commands were executed in a Kali Linux virtual machine (footprinting) and a Windows environment with Zenmap installed (network scanning). Every step below documents the exact command or scan profile used, the results observed, evidentiary outputs, and an analysis from a defensive and attacker perspective.

3. Tools Used

The table below lists each tool utilized during the reconnaissance and network discovery phases and its specific purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used to execute OSINT footprinting tools and local network discovery scans.
theHarvester (v4.10.1)	Passive Open Source Intelligence (OSINT) tool used to gather emails, subdomains, hosts, IPs, and employee names from public engines and databases.
Baidu Module (-b baidu)	Search engine plugin within theHarvester used to query indexed web records without requiring third-party API credentials.
Zenmap (Nmap GUI)	Graphical network scanner used to sweep the local subnet, identify active live hosts, inspect IP details, and generate a visual network topology.
Windows CMD (ipconfig)	Command-line utility used to determine local IPv4 address and subnet configuration prior to scanning.

4. Activities Performed

4.1 Footprinting & Reconnaissance with theHarvester

I conducted passive OSINT reconnaissance against microsoft.com using theHarvester in Kali Linux across two test runs to evaluate both multi-source enumeration and targeted engine querying.

Test Run 1 — Broad-Spectrum Multi-Source Scan (-b all): First, I executed theHarvester -d microsoft.com -l 50 -b all to query all integrated data sources with a limit of 50 results. The tool loaded proxy settings from /etc/theHarvester/proxies.yaml and attempted to query modules including BeVigil, Bitbucket, BuiltWith, Censys, CriminalIP, Dehashed, DNSDumpster, FullHunt, GitHub, HaveIBeenPwned, Hunter, IntelX, Netlas, Onyphe, SecurityTrails, Shodan, VirusTotal, WhoisXML, and ZoomEye.

Because third-party API keys were not configured in /etc/theHarvester/api-keys.yaml, authenticated modules returned [!] Missing API key warnings, while several unauthenticated endpoints (such as Leak-Lookup and THC) experienced DNS timeout exceptions. Consequently, the broad scan returned 0 hosts, 0 IPs, 0 emails, and 0 LinkedIn users.

Test Run 2 — Targeted Search Engine Query (-b baidu): Next, I executed theHarvester -d microsoft.com -l 1000 -b baidu to test an unauthenticated search source with a result limit of 1000. The tool initialized cleanly, queried Baidu for indexed records associated with microsoft.com, and completed with 0 IPs, 0 emails, 0 people, and 0 hosts exposed.

4.2 Network Scanning with Zenmap

For the second practical activity, I used Zenmap to perform active host discovery on my local area network (LAN). The objective was to identify my local subnet, discover live hosts, inspect host details, and generate a visual network topology.

I first verified my local IPv4 configuration and subnet range (10.0.0.0/24). I then entered 10.0.0.0/24 as the target in Zenmap and selected the Ping scan profile, which executes the non-intrusive host discovery command nmap -sn 10.0.0.0/24 across all 256 IP addresses in the subnet.

My scan identified 2 live hosts active on the local network:

- 10.0.0.1
- 10.0.0.2

After the Ping scan completed, I opened the Topology tab in Zenmap, enabled the Legend panel to interpret host icons and connection paths, and saved the network topology diagram for documentation.

5. Risk Analysis / Impact

Based on the observations gathered during the footprinting and local network scanning exercises, the following security considerations and potential risks were identified:

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	OSINT Tool Dependency on API Credentials	theHarvester -d microsoft.com -l 50 -b all returned missing API key errors for Shodan, Censys, VirusTotal, and others.	Without configured API keys in api-keys.yaml, automated OSINT assessments miss deeper infrastructure and threat intelligence data.	Low
2	Minimal Public Exposure on Target Domain	Targeted Baidu module scan (-b baidu) returned 0 emails, 0 IPs, and 0 subdomains for microsoft.com.	Enterprise domains enforce strict perimeter controls and public data scrubbing, preventing trivial passive enumeration via standard search engines.	Low

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
3	Live Hosts Discoverable on Local Subnet	Zenmap Ping Scan (nmap -sn 10.0.0.0/24) identified 2 active hosts (10.0.0.1 and 10.0.0.2) on the LAN.	Any unauthorized device connected to the local network can enumerate live internal IP addresses and map network topology for further targeted probing.	Medium

Risk Level Key: Critical | High | Medium | Low
Note: The risks above represent passive reconnaissance and basic host discovery observations, not confirmed vulnerabilities. No exploitation or unauthorized access was performed during these exercises.

6. Recommendations

Based on the findings from these activities, I recommend the following security and operational improvements:

1. Configure Authorized API Integrations: Security analysts using theHarvester should populate valid API keys inside /etc/theHarvester/api-keys.yaml to unlock full visibility across passive DNS and threat intelligence providers.
2. Perform Regular Passive Footprinting Audits: Organizations should periodically run OSINT tools against their own public domains to verify that employee email formats, internal hostnames, and technical metadata are not unintentionally indexed.
3. Enforce Data Leak Prevention (DLP): Scrub public-facing documents, repositories, and web headers to minimize unnecessary technology or personnel exposure.
4. Conduct Periodic Internal Network Discovery: Administrators should routinely scan internal subnets (such as 10.0.0.0/24) to verify active hosts (10.0.0.1 and 10.0.0.2) against authorized asset inventories.
5. Investigate Unknown Devices Immediately: Any unrecognized IP or MAC address discovered during internal Ping scans should be isolated and investigated.
6. Maintain Updated Network Topology Documentation: Keep network maps and subnet inventories current so unauthorized changes are detected promptly.
7. Operate Strictly Within Authorized Scope: Reconnaissance and active scanning must only be executed against systems and networks with explicit written permission.

7. Conclusion

During Week 2 of my Cybersecurity internship at Networkwalks (Batch BO84), I completed practical exercises covering passive OSINT footprinting with theHarvester on Kali Linux and active network discovery with Zenmap.

In the footprinting activity, I learned how theHarvester queries multiple public sources, how API key configuration in api-keys.yaml governs intelligence depth, and how hardened enterprise targets like microsoft.com minimize passive search exposure. In the network scanning activity, I scanned my local 10.0.0.0/24 subnet using a Ping scan (nmap -sn), discovered 2 active hosts (10.0.0.1 and 10.0.0.2), and generated a visual network topology.

These exercises reinforced that structured reconnaissance and accurate technical documentation—linking commands, evidence, risk analysis, and remediation—are foundational skills in ethical hacking and defensive cybersecurity.